

EXERCISE 06: The Hebb's rule and the perceptron rule

Consider a student perceptron with N binary input elements and one binary output receiving from an identical teacher perceptron the classification results for $P = \alpha N$ random examples $\{\tilde{\xi}^\alpha\}$ of a training set.

- Apply the Hebb's rule to evaluate $\vec{J}^H$.
- Repeat the procedure 1000 times in correspondence of different training sets and different teacher perceptrons and evaluate as a function of α the average training error and the average test error on a test set with $P_{test} = 1000$.
- Superimpose the analytic predictions

$$\epsilon(\alpha) = \frac{1}{\pi} \arccos \left(\sqrt{\frac{2\alpha}{2\alpha + \pi}} \right)$$

and

$$\epsilon_{train}(\alpha) = 2 \int_0^\infty Dx H \left(\frac{1}{\sqrt{\alpha}} + \sqrt{\frac{2\alpha}{\pi}} x \right)$$

with

$$H(u) = \int_u^\infty Dy$$

and

$$Dy = \frac{dy}{\sqrt{2\pi}} \exp(-y^2/2) .$$

- Focus in particular on their small and large α behaviour and the behaviour of $\epsilon - \epsilon_{train}$ at large α .
- Compare them with the results of the randomised perceptron rule used in Exercise 03 and the predictions from Gibbs learning and discuss what you obtain.
- Apply the perceptron rule (without any added noise) to evaluate $\vec{J}^*$.
- Repeat the procedure 1000 times in correspondence of different training sets and different teacher perceptrons and evaluate as a function of α the average test error on a test set with $P_{test} = 1000$.
- Compare them with the results of the randomised perceptron rule used in Exercise 03 and the predictions from Gibbs learning and discuss what you obtain.